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### Field effect transistor with backside source/drain

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#### Abstract

A semiconductor device includes a field effect transistor (FET). The FET includes a gate and a first source or drain (S/D) region. A frontside S/D contact may be connected to and extend vertically upward from a top surface of the first S/D region. The FET further includes a second S/D region. The second S/D region includes a conduit liner and an inner column internal to the conduit liner that extends below a bottom surface of the wraparound gate. A backside S/D contact may be connected to and extend vertically downward from a bottom surface of the second S/D region.

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## Background/Summary

### BACKGROUND

(1) The present disclosure relates to fabrication methods and resulting structures for semiconductor devices. More specifically, the present disclosure relates to fabrication methods and resulting structures for field effect transistors (FETs) with a backside source/drain (S/D) for complementary metal oxide semiconductor (CMOS) technologies.

(2) In certain semiconductor device fabrication processes, many semiconductor devices, such as n-type field effect transistors (nFETs) and p-type field effect transistors (pFETs), may be fabricated on a single wafer. Non-planar transistor device architectures (e.g., fin-type FETs (FinFETs), gate all around (GAA) FETs, nanowire FETs, nanosheet FETs, or the like) can provide increased device density and increased performance over planar transistors. As semiconductor integrated circuits (ICs) and/or chips become smaller, the implementation of stacked nanostructure channels in semiconductor devices has increased. Nanosheets or nanowires generally refer to two-dimensional nanostructures with a thickness range on the order of about 1 nanometer (nm) to about 100 nm, and they can serve as FET channels and facilitate the fabrication of non-planar semiconductor devices having a reduced footprint compared to conventional planar-type semiconductor devices. For example, nanosheet transistors, in contrast to conventional planar FETs, include a gate stack that wraps around the full perimeter of multiple stacked nanosheet channels for a reduced device footprint and improved control of channel current flow. Nanosheet transistor configurations may enable fuller depletion in the nanosheet channel regions and reduce short-channel effects.

Accordingly, nanosheets and nanowires are seen as feasible options for CMOS technology at 3 nm node and beyond.

### SUMMARY

(3) In an embodiment of the present disclosure, a semiconductor device is presented. The semiconductor device includes an insulating layer and a transistor upon the insulating layer. The transistor includes one or more channel regions, a first source or drain (S/D) region, and a second S/D region. The first S/D region is located upon the insulating layer and is connected to the one or more channel regions. The second S/D region includes a conduit liner upon the insulating layer. The conduit liner is connected to the one or more regions. The second S/D region further includes an inner column within the conduit liner. The inner column extends below a top surface of the insulating layer.

(4) In an embodiment of the present disclosure, a method of forming a semiconductor device is presented. The method includes forming a first source or drain (S/D) region upon an insulating layer and contacting respective first ends of one or more channel regions. The method further includes forming a second S/D region upon the insulating layer and contacting respective second ends of the one or more channel regions. The method further includes forming a S/D conduit liner by forming an inner opening through the second S/D region. The inner opening extends below an upper surface of the insulating layer. The method further includes forming a S/D inner column within the inner opening.

(5) In an embodiment of the present disclosure, a transistor is presented. The transistor includes a one or more channel regions and one gate around each of the one or more channel regions. The transistor further includes a first source or drain (S/D) region connected to each of the one or more channel regions. The transistor further includes a second S/D region. The second S/D region includes a conduit liner connected to each of the one or more channel regions and an inner column internal to the conduit liner that extends below a bottom surface of the gate.

(6) The above summary is not intended to describe each illustrated embodiment or every implementation of the present disclosure.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawings included in the present application are incorporated into, and form part of, the specification. They illustrate embodiments of the present disclosure and, along with the description, explain the principles of the disclosure. The drawings are only illustrative of certain embodiments and do not limit the disclosure.
- (2) FIG. 1 depicts a partial top-down view and cross-sectional views of a semiconductor device that includes a backside S/D at an intermediate stage of a fabrication process, according to embodiments.
- (3) FIG. 2 through FIG. 5 depict cross-sectional views of a semiconductor device that includes a backside S/D at intermediate stages of the fabrication process, according to embodiments.
- (4) FIG. 6 and FIG. 7 depict partial top-down views and cross-sectional views of a semiconductor device that includes a backside S/D at intermediate stages of the fabrication process, according to embodiments.
- (5) FIG. 8 through FIG. 13 depict cross-sectional views of a semiconductor device that includes a backside S/D at intermediate stages of the fabrication process, according to embodiments.
- (6) FIG. 14 depicts a process of fabricating a semiconductor device that includes a backside S/D, according to embodiments.

### DETAILED DESCRIPTION

- (7) The present disclosure describes nanostructure FET devices, such as a nanosheet GAA FET, and methods of fabrication the FET devices. In particular, the present disclosure describes a nanostructure FET that includes a backside S/D. In certain examples, the backside S/D is associated with an internal epitaxially grown S/D.
- (8) The flowcharts and cross-sectional diagrams in the Figures illustrate methods of fabricating FET devices, according to various embodiments. In some alternative implementations, the fabrication steps may occur in a different order that that which is noted in the Figures, and certain additional fabrication steps may be implemented between the steps noted in the Figures. Moreover, any of the layered structures depicted in the Figures may contain multiple sublayers.
- (9) Various embodiments of the present disclosure are described herein with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of the present disclosure. It is noted that various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the present disclosure is not intended to be limiting in this respect. Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship. As an example of an indirect positional relationship, references in the present description to forming layer “A” over layer “B” include situations in which one or more intermediate layers (e.g., layer “C”) is between layer “A” and layer “B” as long as the relevant characteristics and functionalities of layer “A” and layer “B” are not substantially changed by the intermediate layer(s).
- (10) The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only

those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

(11) For purposes of the description hereinafter, the terms “upper,” “lower,” “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” and derivatives thereof shall relate to the described structures and methods, as oriented in the drawing figures. The terms “overlying,” “atop,” “on top,” “positioned on” or “positioned atop” mean that a first element, such as a first structure, is present on a second element, such as a second structure, wherein intervening elements such as an interface structure can be present between the first element and the second element. The term “direct contact” means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary conducting, insulating or semiconductor layers at the interface of the two elements. It should be noted, the term “selective to,” such as, for example, “a first element selective to a second element,” means that a first element can be etched, and the second element can act as an etch stop.

(12) For the sake of brevity, conventional techniques related to semiconductor device and IC fabrication may or may not be described in detail herein. Moreover, the various tasks and process steps described herein can be incorporated into a more comprehensive procedure or process having additional steps or functionality not described in detail herein. Various steps in the manufacture of semiconductor devices and semiconductor-based ICs are well known and so, in the interest of brevity, many conventional steps will only be mentioned briefly herein or will be omitted entirely without providing the well-known process details.

(13) In general, the various processes used to form a micro-chip that will be packaged into an IC fall into four general categories, namely, film deposition, removal/etching, semiconductor doping and patterning/lithography. Deposition is any process that grows, coats, or otherwise transfers a material onto the wafer. Available technologies include physical vapor deposition (PVD), chemical vapor deposition (CVD), electrochemical deposition (ECD), molecular beam epitaxy (MBE) and more recently, atomic layer deposition (ALD) among others. Removal/etching is any process that removes material from the wafer. Examples include etch processes (either wet or dry), and chemical-mechanical planarization (CMP), and the like. Semiconductor doping is the modification of electrical properties by doping, for example, transistor sources and drains, generally by diffusion and/or by ion implantation. These doping processes are followed by furnace annealing or by rapid thermal annealing (RTA). Annealing serves to activate the implanted dopants. Films of both conductors (e.g., poly-silicon, aluminum, copper, etc.) and insulators (e.g., various forms of silicon dioxide, silicon nitride, etc.) are used to connect and isolate transistors and their components. Selective doping of various regions of the semiconductor substrate allows the conductivity of the substrate to be changed with the application of voltage. By creating structures of these various components, millions of transistors can be built and wired together to form the complex circuitry of a modern microelectronic device. Semiconductor lithography is the formation of three-dimensional relief images or patterns on the semiconductor substrate for subsequent transfer of the pattern to the substrate. In semiconductor lithography, the patterns are formed by a light sensitive polymer called a photo-resist. To build the complex structures that make up a transistor and the many wires that connect the millions of transistors of a circuit, lithography and etch pattern transfer steps are repeated multiple times. Each pattern being printed on the wafer is aligned to the previously formed patterns and slowly the conductors, insulators and selectively doped regions are built up to form the final device.

(14) Turning now to an overview of technologies that are more specifically relevant to aspects of the present disclosure, a metal-oxide-semiconductor field-effect transistor (MOSFET) may be used for amplifying or switching electronic signals. The MOSFET has a source electrode, a drain electrode, and a metal oxide gate electrode. The metal gate portion of the metal oxide gate electrode is electrically insulated from the main semiconductor n-channel or p-channel by a thin layer of insulating material, for example, silicon dioxide or high-k dielectric, which makes the

input resistance of the MOSFET relatively high. The gate voltage controls whether the current path from the source to the drain is an open circuit (“off”) or a resistive path (“on”). N-type field effect transistors (nFET) and p-type field effect transistors (pFET) are two types of complementary MOSFETs. The nFET includes n-doped source and drain junctions and uses electrons as the current carriers. The pFET includes p-doped source and drain junctions and uses holes as the current carriers. Complementary metal oxide semiconductor (CMOS) is a technology that uses complementary and symmetrical pairs of p-type and n-type MOSFETs to implement logic functions. As mentioned above, hole mobility on the pFET may have an impact on overall device performance.

(15) The wafer footprint of an FET is related to the electrical conductivity of the channel material. If the channel material has a relatively high conductivity, the FET can be made with a correspondingly smaller wafer footprint. A known method of increasing channel conductivity and decreasing FET size is to form the channel as a nanostructure. For example, a GAA nanosheet FET provides a relatively small FET footprint by forming the channel region as a series of nanosheets. In a known GAA configuration, a nanosheet GAA FET includes a source region, a drain region and stacked nanosheet channels between the source and drain regions. Semiconductor nanosheet FET devices typically include one or more suspended nanosheets that serve as the channel. A gate surrounds the stacked nanosheet channels and regulates electron flow through the nanosheet channels between the source and drain regions. GAA nanosheet FETs are fabricated by forming alternating layers of channel nanosheets and sacrificial nanosheets. The sacrificial nanosheets are released from the channel nanosheets before the FET device is finalized. For n-type FETs, the channel nanosheets are typically silicon (Si) and the sacrificial nanosheets are typically silicon germanium (SiGe). For p-type FETs, the channel nanosheets can be SiGe and the sacrificial nanosheets can be Si. In some implementations, the channel nanosheet of a p-type FET can be SiGe or Si, and the sacrificial nanosheets can be Si or SiGe. Forming the GAA nanosheets from alternating layers of channel nanosheets formed from a first type of semiconductor material (e.g., Si for n-type FETs, and SiGe for p-type FETs) and sacrificial nanosheets formed from a second type of semiconductor material (e.g., SiGe for n-type FETs, and Si for p-type FETs) provides superior channel electrostatics control, which is necessary for continuously scaling gate lengths down to seven (7) nanometer CMOS technology and below.

(16) In certain nanostructure FET devices, it has been difficult to integrate a backside S/D contact to contact the S/D from the backside of the semiconductor device. For example, in known nanostructure FET devices, a sacrificial S/D contact is initially formed prior to forming an associated S/D thereupon. Subsequently, the sacrificial S/D contact, may be removed and replaced with a backside S/D contact. The removal of the sacrificial S/D contact is difficult and increases semiconductor device fabrication complexity. Therefore, it may be desirable to fabricate nanostructure FET structures with a backside S/D and/or an associated backside S/D contact without an associated sacrificial S/D contact.

(17) Referring now to the drawings in which like numerals represent the same or similar elements and initially to FIG. 1, this figure depicts a top-down view and cross-sectional views of a semiconductor device **100** at an intermediate stage of the fabrication process, according to embodiments.

(18) The semiconductor device **100** may be formed over a substrate structure. The substrate structure may be a bulk-semiconductor substrate. In one example, the bulk-semiconductor substrate may be a silicon-containing material. Illustrative examples of silicon-containing materials suitable for the bulk-semiconductor substrate include, but are not limited to, silicon, silicon germanium, silicon germanium carbide, silicon carbide, polysilicon, epitaxial silicon, amorphous silicon, and multi-layers thereof. Although silicon (Si) is the predominantly used semiconductor material in wafer fabrication, alternative semiconductor materials can be employed, such as, but not limited to, gallium arsenide, gallium nitride, cadmium telluride, zinc selenide, and III-V compound

semiconductors and/or II-VI compound semiconductors. III-V compound semiconductors are materials that include at least one element from Group III of the Periodic Table of Elements and at least one element from Group V of the Periodic Table of Elements. II-VI compound semiconductors are materials that include at least one element from Group II of the Periodic Table of Elements and at least one element from Group VI of the Periodic Table of Elements. In another implementation, as depicted, the substrate structure includes a substrate **102** and an insulator layer **104**. The substrate **102** may be comprised of any other suitable material(s) that those listed above and the insulator layer **104** may be a dielectric layer, such as an oxide, and may be referred to as a buried oxide (BOX) layer. The dielectric layer may be any suitable dielectric, oxide, or the like, and it may electrically isolate the nanostructures from the bottom substrate **102**.

(19) As shown in FIG. 1, after initial fabrication processing, semiconductor device **100** may include a substrate **102**, an insulator layer **104**, a nanosheet stack that includes semiconductor layers **108** and sacrificial layers **106**, a sacrificial gate **116**, a gate hardmask **118**, a gate spacer **120**, and inner spacers **122**.

(20) The insulator layer **104** may be formed on the substrate **102**. Thus, in various examples, substrate **102** is provided, insulator layer **104** is deposited over substrate **102**, and then the nanosheet stack is formed over the insulator layer **104**. Alternatively, the initial substrate may be an insulator on substrate, such as a SiGeOI (SiGe on insulator substrate), a SOI (Silicon on insulator substrate, or the like).

(21) The nanosheet stack may be formed by forming alternating sacrificial layers and active semiconductor layers. In certain examples, the first one of the sacrificial layers is initially formed directly on an upper surface of the insulator layer **104**. In other examples, certain layers may be formed between the upper surface of the insulator layer **104** and the first one of the sacrificial layers. In an example, the sacrificial layer is composed of silicon-germanium (e.g., SiGe, where the Ge ranges from about 25-40%). Next, an active semiconductor layer is formed on an upper surface of the first one of the sacrificial layers. In an example, the active semiconductor layer is composed of silicon. Several additional layers of the sacrificial layer and the active semiconductor layer are alternately formed. In the example illustrated, there are a total of three sacrificial layers **106** and three active semiconductor layers **108** that are alternately formed to form the nanosheet stack. However, it should be appreciated that any suitable number of alternating layers may be formed. Although it is specifically contemplated that the sacrificial layers can be formed from SiGe and that the active semiconductor layers can be formed from Si, it should be understood that any appropriate materials can be used instead, as long as the two semiconductor materials have etch selectivity with respect to one another. As used herein, the term “selective” in reference to a material removal process denotes that the rate of material removal for a first material is greater than the rate of removal for at least another material of the structure to which the material removal process is being applied. The alternating semiconductor materials can be deposited by any appropriate mechanism. It is specifically contemplated that the first and second semiconductor materials (i.e., of the sacrificial layers and the active semiconductor layers) can be epitaxially grown from one another, but alternate deposition processes, such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or gas cluster ion beam (GCIB) deposition, are also contemplated.

(22) In certain embodiments, the sacrificial layers have a vertical thickness ranging, for example, from approximately 3 nm to approximately 20 nm. In certain embodiments, the active semiconductor layers have a vertical thickness ranging, for example, from approximately 3 nm to approximately 10 nm. Although six total layers are associated with the depicted semiconductor structure **100**, it should be appreciated that the nanosheet stack can include any suitable number of layers. Although the range of 3-20 nm is cited as an example range of thickness, other thicknesses of these layers may be used. In certain examples, certain of the sacrificial layers or the active semiconductor layers may have different thicknesses relative to one another. Therefore, multiple

epitaxial growth processes can be performed to form the alternating sacrificial layers and the active semiconductor layers.

(23) In certain embodiments, it may be desirable to have a small vertical spacing (VSP) between adjacent nanosheet layers in a stack of nanosheets to reduce the parasitic capacitance and to improve circuit speed. For example, the VSP (the distance between the bottom surface of a first nanosheet layer and the top surface of an adjacent second nanosheet layer) may range from 5 nm to 15 nm. However, the VSP must be of a sufficient value to accommodate the gate stack that will be formed in the spaces created by later removal of the sacrificial layers **106**.

(24) In some implementations, a mask layer (not shown) is formed on the uppermost active semiconductor layer. The mask layer may be comprised of any suitable material(s) known to one of skill in the art. The mask layer is patterned and used to perform the nanosheet patterning process. In the nanosheet patterning process, any suitable material removal process (e.g., reactive ion etching or RIE) may be used to remove the various layers of the nanosheet stack down to the level of the insulator layer **104**. Following the patterning process to the nanosheet stacks, thereby forming sacrificial layers **106** and semiconductor layers **108**, respectively, the mask is removed.

(25) Subsequently, a sacrificial gate **116** is formed on the sacrificial gate oxide layer by any suitable deposition and/or patterning processes known to one of skill in the art. In one example, the sacrificial gate **116** is formed by depositing a thin SiO<sub>2</sub> sacrificial gate oxide layer, followed by depositing a layer of amorphous silicon (a-Si) as the sacrificial gate **116**. The sacrificial gate **116** may be composed of polycrystalline silicon (poly silicon), amorphous silicon, and/or an oxide, such as, SiO<sub>2</sub>. A gate hardmask **118** is also formed on a top side of the sacrificial gate **116**. The gate hardmask **118** is formed for subsequent nanosheet patterning. The gate hardmask **118** can be composed of various nitride materials including, but not limited to, a nitride, an oxide, silicon nitride (SiN), and/or a combination of a nitride material and an oxide material. In certain embodiments the sacrificial gate **116** extends into and out of the page to wrap around the edges of the nanosheet stack, and the subsequent removal of the sacrificial gate **116** allows an access point for later removal of the sacrificial layers. In certain examples, gate patterning may be performed by first patterning the gate hardmask **118** and then using the patterned gate hardmask **118** to etch the sacrificial gates **116**. For clarity, the combined structure of the sacrificial gate **116** and gate hardmask **118** may be referred to herein as a sacrificial gate structure **121**.

(26) Subsequently, spacer **120** (or spacer layer) is formed on the sidewalls of the patterned sacrificial gate **116** and the gate hardmask **118**. In certain examples, the spacer **120** is also formed over the sacrificial oxide layer.

(27) Subsequently, the semiconductor device **100** is subjected to a directional reactive ion etch (RIE) process, which can remove portions of the sacrificial layers **106** not covered by the sacrificial gate **116** (and the sacrificial gate hardmask **118**) and the spacer **120**. The RIE can use a boron-based chemistry or a chlorine-based chemistry, for example, which selectively recesses the exposed portions sacrificial layers **106** without significantly removing the active semiconductor layers **108**.

(28) Subsequently, inner spacers **122** are added in the recesses that were previously formed into the sacrificial layers **106**. In certain embodiments, after the formation of the inner spacers **122**, an isotropic etch process is performed to create outer vertical edges to the inner spacers **122** that align with outer vertical edges of the active semiconductor layers **108**. In certain embodiments, the material of the inner spacer **122** is a dielectric material such as SiN, SiO, SiBCN, SiOCN, SiCO, etc.

(29) Referring now to FIG. 2, this figure depicts cross-sectional views of the semiconductor device **100** after additional fabrication operations where source/drain (S/D) regions **124** are formed over the sidewalls of neighboring nanosheet stacks and over the insulator layer **104** and where interlayer dielectric (ILD) **126** is formed over the S/D regions **124** and over the insulator layer **104**.

(30) The S/D region **124** forms either a source or a drain, respectively, to each neighboring nanostructure stacks. S/D regions **124** may be epitaxially grown or formed. The terms “epitaxial



growth and/or deposition” and “epitaxially formed and/or grown” mean the growth of a semiconductor material (crystalline material) on a deposition surface of another semiconductor material (crystalline material), in which the semiconductor material being grown (crystalline overlayer) has substantially the same crystalline characteristics as the semiconductor material of the deposition surface (seed material). In an epitaxial deposition process, the chemical reactants provided by the source gases are controlled and the system parameters are set so that the depositing atoms arrive at the deposition surface of the semiconductor substrate with sufficient energy to move about on the surface such that the depositing atoms orient themselves to the crystal arrangement of the atoms of the deposition surface. Therefore, an epitaxially grown semiconductor material has substantially the same crystalline characteristics as the deposition surface on which the epitaxially grown material is formed. For example, an epitaxially grown semiconductor material deposited on a (100) orientated crystalline surface will take on a (100) orientation. In some embodiments, epitaxial growth and/or deposition processes are selective to forming on semiconductor surfaces, and generally do not deposit material on exposed surfaces, such as silicon dioxide or silicon nitride surfaces.

(31) The S/D region **124** may be formed by epitaxially growing a source/drain epitaxial region within the recess or opening between neighboring nanostructure stacks. In some examples, S/D region **124** is formed by in-situ doped epitaxial growth. In some embodiments, the S/D region **124** epitaxial growth may overgrow above the upper surface of the semiconductor device **100**.

(32) Suitable n-type dopants include but are not limited to phosphorous (P), and suitable p-type dopants include but are not limited to boron (B). The use of an in-situ doping process is merely an example. For instance, one may instead employ an ex-situ process to introduce dopants into the source and drains. Other doping techniques can be used to incorporate dopants in the bottom source/drain region. Dopant techniques include but are not limited to, ion implantation, gas phase doping, plasma doping, plasma immersion ion implantation, cluster doping, infusion doping, liquid phase doping, solid phase doping, in-situ epitaxy growth, or any suitable combination of those techniques. In preferred embodiments, the S/D epitaxial growth conditions that promote in-situ Boron doped SiGe for p-type transistor and phosphorus or arsenic doped silicon or Si:C for n-type transistors. The doping concentration in the S/D region **124** can be in the range of  $1 \times 10^{19} \text{ cm}^{-3}$  to  $2 \times 10^{21} \text{ cm}^{-3}$ , or preferably between  $2 \times 10^{20} \text{ cm}^{-3}$  to  $7 \times 10^{20} \text{ cm}^{-3}$ .

(33) In certain implementations, the S/D region **124** may be partially recessed such that an upper portion of the S/D region **124** are removed. For example, the upper portion of the one or more S/D regions **124** may be etched or otherwise removed. The etch may be timed or otherwise controlled to stop the removal of the one or more S/D regions **124** such that the top surface of the remaining one or more S/D regions **124** is above the upper surface of the topmost active semiconductor layer **108**.

(34) The ILD **126** may be formed on the one or more S/D regions **124**. The ILD **126** may be formed by depositing a dielectric material, upon S/D regions **124** and upon insulator layer **104**. The ILD **126** can be any suitable material, such as, for example, porous silicates, carbon doped oxides, silicon dioxides, silicon nitrides, silicon oxynitrides, or other dielectric materials. Any known manner of forming the ILD **126** can be utilized. The ILD **126** can be formed using, for example, CVD, PECVD, ALD, flowable CVD, spin-on dielectrics, or PVD.

(35) In an example, the ILD **126** may be formed to a thickness above the top surface of the semiconductor device **100** and subsequently etched back such that the top surface of the ILD **126** is coplanar with a top surface of the sacrificial gate structure **121** and/or a top surface of spacer(s) **120**. In another example, another planarization process, such as a CMP, may be performed to create a planar surface for the semiconductor device **100**.

(36) Referring now to FIG. 3, this figure depicts cross-sectional views of the semiconductor device **100** after additional fabrication operations where a mask **140** is formed over ILD, over the sacrificial gate structure **121** and/or over spacer **120**, mask **140** is patterned creating an opening

therein over a second S/D region **124** which will be electrically connected to backside of the wafer in subsequent processes. The mask **140** can be a photolithography mask, hardmask, or the like, and may be composed of various nitride materials including, but not limited to, on OPL, a nitride, an oxide, silicon nitride (SiN), and/or a combination of a nitride material and an oxide material. In certain examples, mask **140** patterning may be performed by conventional lithography and etch process and then using the patterned gate mask **140** to etch or remove portions of ILD **126**, the second S/D region **124**, and insulator layer **104** to form backside S/D trench **142**, as depicted in FIG. **4**. For clarity, patterning of mask **140** may form an opening therein with a central axis **125** that is inline or internal to a perimeter or footprint of the second S/D region **124**. In one example, as depicted, central axis **125** may be coincident or substantially coincident with a center axis, bisector, or the like, of the second S/D region **124**.

(37) Referring now to FIG. **4**, this figure depicts cross-sectional views of the semiconductor device **100** after additional fabrication operations where backside S/D trench **142** is formed through the ILD **126**, the second S/D **124**, and insulator layer **104** generally below the opening within mask **140**. For example, the etching can include a dry etching process in which one or more chemical etchants are used to remove portions of ILD **126**, the second S/D **124**, and insulator layer **104** that are not protected by the patterned mask **140**. As shown in FIG. **4**, in some examples, a perimeter portion of the second S/D region **124** may be retained. In such examples, retention of perimeter portion of the second S/D region **124** may result from the type of etching technique utilized, generally resulting in sloped backside S/D trench **142** sidewalls (as opposed to vertical trench sidewalls). In examples, the top surface of substrate **102** may be utilized as an etch stop, whereby a portion of the top surface **103** of substrate **102** may be exposed by backside S/D trench **142**. The retained perimeter portion of the second S/D region **124** may effectively form a conduit liner or hollow structure (e.g., a hollow column, tube, or the like) that is a full 360-degree perimeter or structure or a partial (not a full 360-degree) perimeter or structure (with reference to the top-down view) to the internal hollow recess therein. For clarity, the conduit liner may be the full 360-degree perimeter or structure (with reference to the top-down view) to the internal hollow recess when the backside S/D trench **142** is vertically aligned with the center of the second S/D region **124** and when the diameter or width dimension(s) of the S/D trench **142** are smaller than the associated second S/D region **124**, respectively. Similarly, the conduit liner may take the form of the partial 360-degree perimeter or structure (with reference to the top-down view) to the internal hollow recess when the backside S/D trench **142** is misaligned with the center of the second S/D region **124** and/or when the diameter or width dimension(s) of the S/D trench **142** is larger than the associated second S/D region **124**.

(38) It should be appreciated that during the removal of ILD **126**, the second S/D region **124**, and insulator layer **104**, appropriate etchants are used, or other etch parameters are selected, to retain the spacers **120**, etc. The etching processes can have etching parameters that can be tuned, such as etchants used, etching temperature, etching solution concentration, etching pressure, source power, RF bias voltage, RF bias power, etchant flow rate, and other suitable parameters. Dry etching processes can include a biased plasma etching process that uses a chlorine-based chemistry. Other dry etchant gasses can include Tetrafluoromethane (CF<sub>4</sub>), nitrogen trifluoride (NF<sub>3</sub>), sulfur hexafluoride (SF<sub>6</sub>), and helium (He), and Chlorine trifluoride (ClF<sub>3</sub>). Dry etching can also be performed anisotropically using such mechanisms as DRIE (deep reactive-ion etching). Chemical vapor etching can be used as a selective etching method, and the etching gas can include hydrogen chloride (HCl), Tetrafluoromethane (CF<sub>4</sub>), and gas mixture with hydrogen (H<sub>2</sub>). Chemical vapor etching can be performed by CVD with suitable pressure and temperature.

(39) Referring now to FIG. **5**, this figure depicts cross-sectional views of the semiconductor device **100** after additional fabrication operations where S/D region **144** is formed within backside S/D trench **142**. The S/D region **144** forms either a source or a drain, respectively, to each neighboring nanostructure stacks. The S/D region **144** may be formed by epitaxially growing a source/drain

epitaxial region within the backside S/D trench **142**. In some examples, S/D region **144** is formed by in-situ doped epitaxial growth. In some examples, the S/D region **144** epitaxial growth may overgrow above the upper surface of the second S/D region **124**. The S/D region **144** may be the same material relative to the second S/D region **124**. Alternatively, the S/D region **144** may be a different material (e.g., different material entirely, different atomic weight of an element of the same material, or the like) relative to the second S/D region **124**. In some examples, S/D region **144** may be epitaxially grown from the top surface **103** of substrate **102** that is exposed by backside S/D trench **142**. As such, a bottom surface of S/D region **144** is generally below a bottom surface of the S/D regions **124**.

(40) For clarity, as depicted, the second S/D structure may resultantly consist of the conduit liner, perimeter shell, tube, or the like, effectively formed by the second S/D region **124** with an inner column (S/D region **144**), therein that, at least, extends below the second S/D region **124** and may also extend above the second S/D region **124**. The inner column or S/D region **144** may directly contact the second S/D region **124**, such that there is no additional electrical impedance between the S/D region **144** and the second S/D region **124**.

(41) In certain implementations, the S/D region **144** may be partially recessed such that an upper portion of the S/D region **144** is removed. For example, the upper portion S/D regions **144** may be etched or otherwise removed. The etch may be timed or otherwise controlled to stop the removal of the S/D regions **144** such that the top surface of the S/D region **144** is below the upper surface of the second S/D region **124**. In these examples, the top surface of S/D region **144** may still be above the upper surface of the topmost active semiconductor layer **108**.

(42) Referring now to FIG. **6**, this figure depicts top-down and cross-sectional views of the semiconductor device **100** after additional fabrication operations where ILD **126** is formed within the remaining backside S/D trench **142**, gate hardmask **118** is removed, and gate cut **150** regions are formed within the sacrificial gate **116**.

(43) The ILD **126** may be formed within the remaining backside S/D trench **142**, generally filling the backside S/D trench **142**. The ILD **126** may be formed by depositing a dielectric material, such as, for example, porous silicates, carbon doped oxides, silicon dioxides, silicon nitrides, silicon oxynitrides, or other dielectric materials. Any known manner of forming the ILD **126** can be utilized. The ILD **126** can be formed using, for example, CVD, PECVD, ALD, flowable CVD, spin-on dielectrics, or PVD.

(44) Gate hardmask **118** may be removed by a planarization process, such as a CMP, that results in a planar upper surface for the semiconductor device **100**. For example, after the planarization process, the top surfaces of spacers **120**, the top surfaces of sacrificial gates **116**, and the top surface of ILD **126** may be coplanar.

(45) Gate cut **150** regions physically and at least partially electrically separates the gate structure (e.g., electrically and physical separates the sacrificial gates **116**). Initially, gate cut **150** regions may be formed by forming a gate cut trench by removing portion(s) of sacrificial gates **116**, gate spacers **120**, within the location of the gate cut region stopping at the top surface of isolation layer **104** or substrate **102**. The undesired portions of these applicable material(s) may be removed removal techniques such as e.g., depositing and patterning a gate cut mask, lithography, etching, or the like.

(46) Subsequently, gate cut **150** region is formed within the gate cut trench. Gate cut region **150** may be a dielectric material, such as, for example, porous silicates, carbon doped oxides, silicon dioxides, silicon nitrides, silicon oxynitrides, or other dielectric materials.

(47) In an example, gate cut **150** region may be connected to and separate a first gate of a p-type nanostructure GAA FET from a second gate of a n-type nanostructure GAA FET. In another example, gate cut **150** region may be connected to and separate a first gate of a p-type nanostructure GAA FET from a second gate of a p-type nanostructure GAA FET. In another example, gate cut **150** region may be connected to and separate a first gate of a n-type

nanostucture GAA FET from a second gate of a n-type nanostucture GAA FET.

(48) Referring now to FIG. 7, this figure is a cross-sectional view of semiconductor device **100** after additional fabrication operations where the sacrificial gate **116** is removed, the sacrificial oxide layer is removed, the sacrificial layers **106** are removed, and replacement gate structure **135** is formed in place thereof.

(49) The sacrificial gate **116** is removed by any suitable material removal process known to one of skill in the art. For example, such removal may be accomplished by an etching process which may include a dry etching process such as, for example, reactive ion etching, plasma etching, ion etching or laser ablation. The etching can further include a wet chemical etching process in which one or more chemical etchants are used to remove portions of the blanket layers that are not protected by the patterned photoresist. Then, the sacrificial layers **106** are removed (or released). Thus, there are void spaces between the active semiconductor layers **108** due to the removal of the sacrificial layers **106**. It should be appreciated that during the removal of the sacrificial gate **116**, the sacrificial oxide layer, and the sacrificial layers **106**, appropriate etchants are used that do not significantly remove material of active semiconductor layers **108**. The dry and wet etching processes can have etching parameters that can be tuned, such as etchants used, etching temperature, etching solution concentration, etching pressure, source power, RF bias voltage, RF bias power, etchant flow rate, and other suitable parameters. Dry etching processes can include a biased plasma etching process that uses a chlorine-based chemistry. Other dry etchant gasses can include Tetrafluoromethane (CF<sub>4</sub>), nitrogen trifluoride (NF<sub>3</sub>), sulfur hexafluoride (SF<sub>6</sub>), and helium (He), and Chlorine trifluoride (ClF<sub>3</sub>). Dry etching can also be performed anisotropically using such mechanisms as DRIE (deep reactive-ion etching). Chemical vapor etching can be used as a selective etching method, and the etching gas can include hydrogen chloride (HCl), Tetrafluoromethane (CF<sub>4</sub>), and gas mixture with hydrogen (H<sub>2</sub>). Chemical vapor etching can be performed by CVD with suitable pressure and temperature.

(50) Replacement gate structure **135** may be formed by initially forming an interfacial layer (not shown) on the interior surfaces of the spacer **120** and the interior surfaces of the active semiconductor layers **108** and the inner spacers **122**. Then, a high-K layer (not shown) is formed to cover the surfaces of exposed surfaces of the interfacial layer **130**. The high-K layer can be deposited by any suitable techniques, such as ALD, CVD, metal-organic CVD (MOCVD), physical vapor deposition (PVD), thermal oxidation, combinations thereof, or other suitable techniques. A high-K dielectric material is a material with a higher dielectric constant than that of SiO<sub>2</sub>, and can include e.g., LaO, AlO, ZrO, TiO, Ta<sub>2</sub>O<sub>5</sub>, Y<sub>2</sub>O<sub>3</sub>, SrTiO<sub>3</sub>(STO), BaTiO<sub>3</sub> (BTO), BaZrO, HfZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfSiO, (Ba,Sr)TiO<sub>3</sub> (BST), Al<sub>2</sub>O<sub>3</sub>, Si<sub>3</sub>N<sub>4</sub>, oxynitrides (SiON), or other suitable materials. The high-K layer can include a single layer or multiple layers, such as metal layer, liner layer, wetting layer, and adhesion layer. In other embodiments, the high-K layer can include, e.g., Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, Al, WN, Cu, W, or any suitable materials.

(51) Replacement gate structure **135** may be further formed by depositing a work function metal (WFM) gate **134** (or replacement gate). The WFM gate **134** can be comprised of metals, such as, e.g., copper (Cu), cobalt (Co), aluminum (Al), platinum (Pt), gold (Au), tungsten (W), titanium (Ti), nitride (N) or any combination thereof. The metal can be deposited by a suitable deposition process, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), physical vapor deposition (PVD), plating, thermal or e-beam evaporation, or sputtering. In various exemplary embodiments, the height of the WFM gate **134** can be reduced by chemical-mechanical polishing (CMP) and/or etching. Therefore, the planarization process can be provided by CMP. Other planarization process can include grinding and polishing. In general, the work function metal (WFM) gate **134** sets the threshold voltage (V<sub>t</sub>) of the device, a high-K gate dielectric material separating the WFM gate **134** from the nanostucture channel (i.e., active

semiconductor layers **108**), and other metals that may be desired to further fine tune the effective work function (eWF) and/or to achieve a desired resistance value associated with current flow through the gate stack in the direction parallel to the plane of the nanostructure channel.

(52) Referring now to FIG. **8**, this figure is a cross-sectional view of semiconductor device **100** after additional fabrication operations, where ILD **126** is formed (e.g., thickened, etc.), S/D contact **160** is formed, BEOL structure(s) **162** are formed, and a carrier wafer **164** is attached.

(53) The ILD **126** may be formed upon the top surface of the semiconductor device **100**, as depicted in FIG. **7**. The ILD **126** may be formed by depositing a dielectric material, such as, for example, porous silicates, carbon doped oxides, silicon dioxides, silicon nitrides, silicon oxynitrides, or other dielectric materials. Any known manner of forming the ILD **126** can be utilized. The ILD **126** can be formed using, for example, CVD, PECVD, ALD, flowable CVD, spin-on dielectrics, or PVD.

(54) Subsequently, S/D contact **160** is formed by forming a S/D contact trench within ILD **126** and depositing conductive material within the S/D contact trench over a first S/D region **124**.

(55) S/D contact **160** may consist of a silicide liner, such as Ni, NiPt or Ti, etc, a metal adhesion liner, such as TiN, TaN, etc., and a conductive metal fill, such as Al, Ru, W, Co, Cu, etc. The formation of S/D contact **160** may include etching the ILD **126** to form via openings, deposition of material(s), and performing a planarization process, such as a CMP process or a mechanical grinding process, to remove excess portions of the conductive barrier layer and the conductive material. Subsequently, the top surface of S/D contact **160** and the top surface of ILD **126** may be coplanar. S/D contact **160** may be formed during middle of the line (MOL) processes.

(56) BEOL structure(s) **162** include metallization levels, associated metallization dielectric layers, VIAs that connect the metallization feature(s) within the metallization levels with an underlying device or structure, and/or conductive bonding pads, or the like. BEOL structure(s) **162** include at least a conductive wire, conductive trace, conductive plane, or the like, that is formed over S/D contact **160** making electrical contact therewith. In some examples, there may be 5 metal levels M.sub.0-M.sub.4 within BEOL structure(s) **162**. In some examples, there may be more than 10 metal levels M.sub.0-M.sub.x within BEOL structure(s) **162**. In some examples, S/D contact **160** may contact and connect the source or drain (i.e., S/D region **124**) of the first nanostructure GAA FET to a trace, power plane, or the like located within the lowest metal level M.sub.0.

(57) Upon completion of BEOL structure(s) **162**, carrier wafer **164** may be bonded or otherwise attached to the top surface or front side of BEOL structure(s) **162**, as depicted. Carrier wafer **164** may be attached to semiconductor device **100** by any carrier bonding technique.

(58) For clarity, semiconductor device **100** may undergo similar processes, utilized to fabricate S/D contact **160**, to form a gate contact, such that the gate contact may contact and extend vertically upward from WFM gate **134** and may connect with a wiring feature within metal levels M.sub.0-M.sub.x within BEOL structure(s) **162**.

(59) Referring now to FIG. **9**, this figure is a cross-sectional view of semiconductor device **100** after additional fabrication operations where the wafer is flipped for backside processing, and then substrate **102** is removed. The substrate **102** may be removed by any removal technique, such as a combination of wafer grinding, CMP, dry and wet etch. Removal of substrate **102** exposes the bottom surface **105** of insulator layer **104** and a bottom surface **145** of S/D region **144**.

(60) Referring now to FIG. **10**, this figure is a cross-sectional view of semiconductor device **100** after additional fabrication operations, where ILD **170** is formed upon the exposed bottom surface **105** of insulator layer **104** and upon the exposed bottom surface **145** of S/D region **144**.

(61) The ILD **170** may be formed by depositing a dielectric material, upon insulator layer **104** and upon S/D region **144**. The ILD **170** can be any suitable material, such as, for example, porous silicates, carbon doped oxides, silicon dioxides, silicon nitrides, silicon oxynitrides, or other dielectric materials. Any known manner of forming the ILD **170** can be utilized. The ILD **170** can be formed using, for example, CVD, PECVD, ALD, flowable CVD, spin-on dielectrics, or PVD.

(62) Referring now to FIG. 11, this figure is a cross-sectional view of semiconductor device **100** after additional fabrication operations, where S/D contact **172** is formed.

(63) Backside S/D contact **172** is formed by forming a S/D contact trench within ILD **170** and depositing conductive material within the S/D contact trench over a S/D region **144**. The S/D contact trench may expose the entire bottom surface **145** of S/D contact **144**. The S/D contact trench may expose a portion of the bottom surface **105** of insulator layer **104** that immediately surrounds the S/D region **144** (e.g., insulator layer **104** is used as an etch stop, etc.). S/D contact **172** may directly contact at least the exposed bottom surface **145** of S/D region **144**. Further, S/D region **172** may directly contact both the exposed bottom surface **145** of S/D region **144** and the exposed portion of the bottom surface **105** of insulator layer **104** that immediately surrounds the S/D region **144**.

(64) The backside S/D contact **172** may consist of a silicide liner, such as Ni, NiPt or Ti, etc., a metal adhesion liner, such as TiN, TaN, etc., and a conductive metal fill, such as Al, Ru, W, Co, Cu, etc. In some examples, S/D contact **172** may be a contact feature, such as a vertical interconnect access (VIA) to a backside rail **180**, shown in FIG. 12, such as a backside wiring line, backside power plane, or the like. The formation of the backside S/D contact **172** may include etching the ILD **170** to form via openings, forming a blanket conductive barrier layer extending into the via openings, depositing a metallic or conductive material over the blanket conductive barrier layer, and performing a planarization process, such as a CMP process or a mechanical grinding process, to remove excess portions of the conductive barrier layer and the conductive material. Subsequently, the bottom surface of S/D contact **172** and the bottom surface of ILD **170** may be coplanar.

(65) In some examples, the thickness of ILD **170** may be chosen to achieve a predetermined adequate dimension between the top surface of S/D contact **172** and WFM gates **134** to prevent or substantially prevent electrical shorting therebetween.

(66) Referring now to FIG. 12, this figure is a cross-sectional view of semiconductor device **100** after additional fabrication operations, where ILD **170** is formed (e.g., thickened, etc.), backside rail **180** is formed, and backside power distribution network (BSPDN) **182** is formed.

(67) The ILD **170** may be formed upon the bottom surface of the semiconductor device **100**, as depicted in FIG. 11. The ILD **170** may be formed by depositing a dielectric material, such as, for example, porous silicates, carbon doped oxides, silicon dioxides, silicon nitrides, silicon oxynitrides, or other dielectric materials. Any known manner of forming the ILD **170** can be utilized. The ILD **170** can be formed using, for example, CVD, PECVD, ALD, flowable CVD, spin-on dielectrics, or PVD.

(68) Subsequently, backside rail **180** is formed by forming a backside rail trench within ILD **170** and depositing conductive material within the backside rail trench over S/D contact **172**. The backside rail trench may expose the entire bottom surface of S/D contact **172** and may further create a surface of ILD **170** that is coplanar with or substantially coplanar with the bottom surface of S/D contact **172**. Backside rail **180** may directly contact at least the exposed bottom surface of S/D contact **172**.

(69) Backside rail **180** may consist of a metal, such as copper, aluminum, tungsten, cobalt, metal alloys thereof, or the like. In some examples, backside rail **180** may be a power plane (e.g., VDD power plane, VSS power plane, etc.). Backside rail **180** may include a conductive region and a conductive barrier layer(s) between the sidewalls and upper surfaces of the conductive regions and the ILD **170**. The conductive barrier layer(s) may be formed of titanium, titanium nitride, tantalum, tantalum nitride, cobalt, a combination thereof, or the like. The conductive regions may be formed of metals such as copper, aluminum, tungsten, cobalt, alloys thereof, or the like. The formation of backside rail **180** may include etching the ILD **170** to form a power plane opening, forming a blanket conductive barrier layer extending into the power plane opening, depositing a metallic or conductive material over the blanket conductive barrier layer, and performing a planarization

process, such as a CMP process or a mechanical grinding process, to remove excess portions of the conductive barrier layer and the conductive material. Subsequently, the bottom surface of backside rail **180** and the bottom surface of ILD **170** may be coplanar. BSPDN **182** includes known power distribution network features and/or structures to adequately provide power to backside rail **180**.

(70) For clarity, as depicted, semiconductor device **100** includes a nanostructure GAA FET with a first region (i.e., source or drain regions such as first S/D region **124**) and associated S/D contact **160** upon a top surface of the first source or drain region. The S/D contact **160** is electrically connected to a BEOL wiring feature generally above the nanostructure GAA FET. The nanostructure GAA FET further includes a second region (i.e., drain or source region such as second S/D region **124**), associated S/D region **144**, and associated S/D contact **172** upon a bottom surface of the associated S/D region **144**. The S/D contact **172** is electrically connected to a backside rail **180** generally below the nanostructure GAA FET.

(71) Alternatively, semiconductor device **100** includes a nanostructure GAA FET with a first region (i.e., source or drain region such as first S/D region **124**), associated first S/D region **144**, and associated first S/D contact **172** upon a bottom surface of the associated first S/D region **144**. The first S/D contact **172** is electrically connected to a first backside rail **180** generally below the nanostructure GAA FET. The nanostructure GAA FET further includes a second region (i.e., drain or source region such as second S/D region **124**), associated second S/D region **144**, and associated second S/D contact **172** upon a bottom surface of the associated second S/D region **144**. The second S/D contact **172** is electrically connected to a second backside rail **180** generally below the nanostructure GAA FET.

(72) As such, in accordance with the embodiments, semiconductor device includes one or more backside S/D contact(s) (i.e., one or more S/D contact(s) **172**) that are respectively electrically connected to a backside power rail **180** and to a S/D region **144**.

(73) Referring now to FIG. **13**, this figure is a cross-sectional view of semiconductor device **100** after alternative fabrication operations where S/D contact **172** is formed. In this alternative fabrication operation, the S/D contact trench within ILD **170** may be formed sequentially. For example, an initial S/D contact trench may be formed by etching with an etch stop set at the may the bottom surface of insulator layer **104**. Next, another etching process may recess S/D region **144**. The etchant chosen may not be entirely selective to the material of insulator layer **104**, which may result in some etching of insulator layer **104** and arced insulator layer **104** sidewalls **173** within the S/D contact trench. These sequential etching operations may result in the dimensions of the S/D contact trench at the ILD **170** and insulator layer **104** planar intersection being greater than the dimensions of the S/D contact trench at the top of the S/D contact trench (i.e., at the bottom surface of the recessed S/D region **144**). The arced sidewalls **173** may have a top-down view radius or diameter that is smallest at the top of the S/D contact **172** and may increase in relation to the depth of the S/D contact **172**, as depicted. In this manner, the S/D contact **172** has a small top-down view radius or diameter upper surface to reduce a likelihood of electrical shorting between the S/D contact **172** and neighboring WFM gate **134** with a relatively larger top-down view radius or diameter lower surface to promote metal or conductive material filling of the associated S/D contact trench.

(74) FIG. **14** depicts a process **200** of fabricating semiconductor device **100** that includes a backside S/D contact **172**, according to embodiments. Process **200** begins at block **202** with patterning a nanostructure stack, forming a sacrificial gate structure around the nanostructure stack, forming a gate spacer around the sacrificial gate structure, directionally recessing the sacrificial nanostructure layers of the nanostructure stack that are under the gate spacer, and forming inner spacers within the recesses formed by the directional spacing. The fabrication operations of block **202** may result in semiconductor device **100** that includes features or structures as depicted in FIG. **1**.

(75) At block **204**, a first S/D region and a second S/D region are formed on either side of the

nanostructure stack. For example, a first S/D region **124** is formed on one sidewall of the nanostructure stack and a second S/D region **124** is formed on the opposing sidewall of the nanostructure stack. In some examples, ILD **126** may be formed upon the first S/D region **124** and the second S/D region **124**. The fabrication operations of block **204** may result in semiconductor device **100** that includes features or structures as depicted in FIG. 2.

(76) At block **206**, a backside opening is formed through the second S/D region. For example, backside S/D trench **142** is formed from a top surface of the semiconductor device **100** (i.e., frontside) through ILD **126**, through the second S/D region **124**, and through insulator layer **104**. The fabrication operations of block **206** may result in semiconductor device **100** that includes features or structures as depicted in FIG. 4.

(77) At block **208**, a third S/D region is formed within the backside opening through the second S/D region. For example, a S/D region **144** is formed within backside S/D trench **142**. The S/D region **144** may directly contact the inner sidewall(s) of the second S/D region **124** that are formed by the backside S/D trench **142**. In some examples, ILD **126** may be formed upon the S/D region **144**. The fabrication operations of block **208** may result in semiconductor device **100** that includes features or structures as depicted in FIG. 5.

(78) At block **210**, the sacrificial gate structure is removed, the sacrificial nanostructure layers within the nanostructure stack are removed, and the active semiconductor layers (i.e., the nanostructure channels) are exposed. For example, sacrificial gate **116** is removed, sacrificial layers **106** are removed, and active semiconductor layers **108** are exposed. At block **212**, a replacement gate structure is formed around the nanostructure channels. For example, WMF gate **134** is formed around the active semiconductor layers **108**. The fabrication operations of blocks **210** and **212** may result in semiconductor device **100** that includes features or structures as depicted in FIG. 7.

(79) At block **214**, a first S/D contact is formed above or over the first S/D region, BEOL structures are formed over the first S/D contact, and a carrier wafer is attached to the top surface or front side of the semiconductor device **100**. For example, S/D contact **160** is formed upon and contacts the top surface of the first S/D region **124**. BEOL structures **162** are formed over the S/D contact **160**, and carrier wafer **164** is attached. The fabrication operations of block **214** may result in semiconductor device **100** that includes features or structures as depicted in FIG. 8.

(80) At block **216**, at least the bottom surface of the third S/D region is exposed from the backside. For example, substrate **102** may be removed and replaced with ILD **170**. A backside S/D contact trench may be formed within ILD **170** that exposes at least the bottom surface of S/D region **144**. The fabrication operations of block **216** may result in semiconductor device **100** that includes features or structures as depicted in FIG. 11.

(81) At block **218**, a second S/D contact is formed below the third S/D region, a backside power rail is formed below the second S/D contact, and/or a BSPSN is formed below the backside power rail. For example, S/D contact **172** may be formed within the backside S/D contact trench, backside rail **180** may be formed upon and below the S/D contact **172**, and BSPDN **182** may be formed upon and below the backside rail **180**. The fabrication operations of block **218** may result in semiconductor device **100** that includes features or structures as depicted in FIG. 12.

(82) The descriptions of the various embodiments have been presented for purposes of illustration and are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

## Claims



1. A semiconductor device comprising: an insulating layer; and a transistor upon the insulating layer, the transistor includes one or more channel regions, a first source or drain (S/D) region upon the insulating layer and connected to the one or more channel regions, and a second S/D region that includes a conduit liner and an inner column, wherein the conduit liner includes a liner around an internal conduit, wherein the liner is upon the insulating layer and is connected to the one or more channel regions, wherein the inner column is composed of a doped semiconductor material and is within the internal conduit of the liner, and wherein the inner column extends below a top surface of the insulating layer.
2. The semiconductor device of claim 1, wherein a bottom surface of the inner column is coplanar with a bottom surface of the insulating layer.
3. The semiconductor device according to claim 1, further comprising: a frontside S/D contact connected to a top surface of the first S/D region.
4. The semiconductor device according to claim 1, further comprising: a backside S/D contact connected to a bottom surface of the inner column.
5. The semiconductor device according to claim 4, further comprising, a backside rail connected to a bottom surface of the backside S/D contact.
6. The semiconductor device of claim 4, wherein a top surface of the backside S/D contact is between the top surface of the insulating layer and a bottom surface of the insulating layer.
7. The semiconductor device according to claim 1, wherein the transistor is a nanosheet gate all around transistor including a plurality of channel regions surrounded by a work function metal gate.
8. The semiconductor device of claim 7, wherein a top surface of the inner column is between a top surface of the work function metal gate and a top surface of the one or more channel regions.
9. The semiconductor device according to claim 1, wherein there is no electrical impedance between the conduit liner and the inner column.
10. The semiconductor device according to claim 1, wherein a top surface of the inner column is above a top surface of the conduit liner.
11. A method of forming a semiconductor device, the method comprising: forming a first source or drain (S/D) region upon an insulating layer and contacting respective first end(s) of one or more channel regions; forming a second S/D region upon the insulating layer and contacting respective second end(s) of the one or more channel regions; forming a S/D conduit liner by forming an inner opening through the second S/D region, wherein the inner opening extends below an upper surface of the insulating layer; and forming a S/D inner column composed of a doped semiconductor material within the inner opening.
12. The method of forming the semiconductor device of claim 11, wherein a bottom surface of the S/D inner column is coplanar with a bottom surface of the insulating layer.
13. The method of forming the semiconductor device of claim 11, further comprising: forming a frontside S/D contact upon a top surface of the first S/D region.
14. The method of forming the semiconductor device of claim 11, further comprising: forming a backside S/D contact upon a bottom surface of the S/D inner column.
15. The method of forming the semiconductor device of claim 14, further comprising: forming a backside rail upon a bottom surface of the backside S/D contact.
16. The method of forming the semiconductor device of claim 11, further comprising: forming a work function metal gate around the one or more channel regions.
17. The method of forming the semiconductor device of claim 16, wherein a top surface of the S/D inner column is between a top surface of the work function metal gate and a top surface of the one or more channel regions.
18. The method of forming the semiconductor device of claim 11, wherein there is no electrical impedance between the conduit liner and the inner column.

19. The method of forming the semiconductor device of claim 11, wherein a top surface of the inner column is above a top surface of the conduit liner.

20. A transistor comprising: one or more channel regions; one gate around each of the one or more channel regions; a first source or drain (S/D) region connected to each of the one or more channel regions; and a second S/D region comprising a conduit liner connected to each of the one or more channel regions and an inner column composed of a doped semiconductor material and internal to the conduit liner that extends below a bottom surface of the gate.

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